# The data at a glance

Start here if prostate cancer or health-records data are new to you. This page describes `data/` and the terms used. It doesn't give answers, and you should still check every file yourself.

## What these data are

The 800 patients are **synthetic**: generated to mimic the structure and recording patterns of real-world health data, with no real individuals. All are men with a confirmed first prostate-cancer diagnosis between 2018-01-01 and 2021-12-31. Median age at diagnosis is 73 years (interquartile range 68–78; range 58–91). The roster holds a patient ID, the diagnosis date and a death date where one was recorded in the registry by 2023-12-31. Each patient then appears in two separate record systems that don't share an ID, which is why a crosswalk file exists.

**Administrative claims** are billing records that providers and pharmacies submit to the payer (the insurer) for reimbursement. Each records a billed service or dispensed prescription and its date. Claims capture care from any provider that bills the payer, but only during enrolment (while the patient is covered). **EHR** (electronic health record) data come from one health system's clinical record. They contain clinical detail such as PSA and tumour grade, but they don't capture care delivered at other institutions. Neither source is complete, and that gap is the point of the exercise.

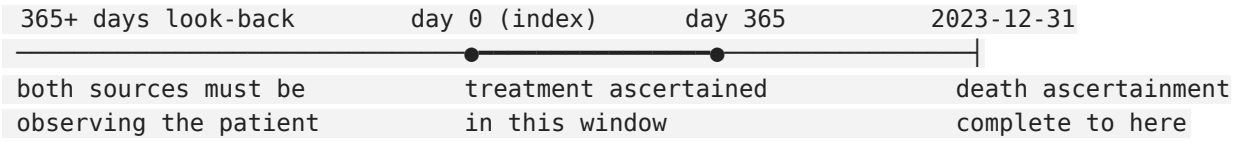
## Key terms

| Term | What it means here |
|---|---|
| Index date | Day of first confirmed diagnosis. Day 0 for every time window. |
| Observation span | The period during which a source captures a patient's care (enrolment for claims, active registration for the EHR). The absence of records outside the span means no data capture, not the absence of clinical events. |
| PSA | Prostate-specific antigen, a serum biomarker reported in ng/mL. Elevated levels are associated with greater tumour burden, but they aren't specific to cancer (they also rise in benign prostatic enlargement and inflammation). PSA tests appear as events but are diagnostic or monitoring tests, not treatment. |
| Grade group | ISUP Grade Group 1–5, a histopathological grade derived from the Gleason score on biopsy tissue. 1 (Gleason ≤6) indicates the most favourable prognosis and 5 (Gleason 9–10) the least favourable. |
| Prostatectomy / radiation | Radical prostatectomy (surgical removal of the prostate and seminal vesicles) and radiotherapy (external-beam radiation or brachytherapy). Both are local treatments directed at the prostate. |
| Systemic therapy | Drug treatment distributed throughout the body via the bloodstream. In these data it's hormonal therapy (androgen-deprivation therapy, ADT, which suppresses testosterone), coded as injectable or oral. No chemotherapy codes are supplied. |
| Part D | The Medicare outpatient prescription-drug benefit. When `part_d = 1`, pharmacy-dispensed oral drugs appear in claims as prescription fills. When it's 0 they're not captured. Injections given in clinic are billed under the medical benefit, so they're visible either way. |
| Registry death date | All-cause date of death from a shared registry, with ascertainment (the recording of every death that occurred) assumed complete through 2023-12-31. |

## What's in each file

| File | Rows | What to know |
|---|---|---|
| `patients.csv` | 800 | The roster. 19% of patients have a recorded death date by 2023-12-31. |
| `patient_linkage.csv` | 800 | Crosswalk tying each shared ID to one claims ID and one EHR ID. |
| `claims_patients.csv` | 800 | Observation span per patient. Most (about 73%) have Part D. Spans start between 2015 and 2021 and run no later than 2023-12-31. |
| `ehr_patients.csv` | 800 | Observation span plus baseline PSA and grade group. Some of each are blank. |
| `claims_events.csv` | about 3,200 | One row per service or fill (ID, date, code), with no event ID, so identical rows can occur. About eight in ten rows are PSA tests. |
| `ehr_events.csv` | about 3,100 | Same layout, for the EHR. Also mostly PSA tests. |
| `code_map.csv` | 10 | Five codes per source: surgery, radiation, injectable ADT, oral ADT and PSA test. All are treatment except the PSA test (a diagnostic event). |

## The timeline




The treatment-capture cohort is smaller than the mortality cohort. Mortality comes from the registry and doesn’t require continued observation in either source. Treatment capture requires both sources to be observing the patient through day 365.

# What changes when we link claims and EHR?

You've got two extracts covering the same 800 synthetic patients. One is insurance claims, the other is records from a single health system. Some treatment shows up in both, and some shows up in only one.

Link them correctly and show how estimated treatment capture (the proportion of patients with a recorded treatment) changes when both sources are used. We'll read your code and plots, and we want to hear what you make of the results.

Plan on 3–4 hours and stop at four, setup and writing included. Use R or Python. Everything you need is supplied, so there's nothing to download or look up. This PDF is the assignment, data guide and README in one, and the only other files are in `data/`.

Don't expect clean data. Files can have errors, missing values and repeated rows. Check the inputs against the definitions, spot the problems and log how you handled them. The appendix describes every column.

## The question

Take the patients who meet the shared observation rules. What proportion have a treatment recorded on days 0–365 after diagnosis in:

1. claims alone,
2. the EHR alone, or
3. either source, once the records are linked?

You'll also describe initial treatment patterns and estimate 365-day all-cause mortality from diagnosis. These two pieces use two separate, clearly defined cohorts. We're testing how you define endpoints and read results. Treatment effectiveness is out of scope.

## What to do

We want to know how much treatment each source captures, whether linkage changes the estimate, and how far you'd trust the result. Choose the statistical methods yourself. We care that they suit the question and that you can say why you picked them.

1. **Define the study populations.** Apply the eligibility and observation rules (page 5) to build two nested cohorts from the roster. Report how many patients are excluded at each step. Be ready to say why the treatment question and the death question need different groups.
2. **Measure treatment in each source.** Among patients observed in both sources through day 365, what share have a treatment recorded in claims, in the EHR, and in either? Give each share with a 95% confidence interval suited to a proportion. Report where the sources agree and disagree, and interpret the differences between the three shares.
3. **Describe first-year treatment patterns.** In the linked records for days 0–365, which treatment modalities (types of treatment) are recorded, alone or in combination, and how many patients have none recorded? Show this in a plot.
4. **Estimate 365-day all-cause mortality.** Define the endpoint, identify the cohort it applies to, and estimate the proportion with an interval. Explain why that group isn't the one in step 2.
5. **Judge how far the data can be trusted.** Say what could make the numbers wrong or incomplete: missing PSA and grade, care outside the EHR's health system, Part D, care in neither extract, and selection effects (how the eligibility criteria change who remains in the cohort). Separate what you can measure from what you can only reason about. Keep a log of the data problems you found and what you did about each.

## Time budget

| Work | Minutes |
|---|---|
| Read the brief and jot a few lines of plan | 20 |
| Join, check the data, log issues, build the cohort | 55 |
| Capture, patterns and the clinical endpoint | 50 |
| Make and check the two plots | 35 |
| Write the report and rerun everything | 40 |
| Buffer | 10 |
| **Target** | **210 (3½ hours)** |

If something's unfinished at four hours, say what it is and what you'd do next. We'd much rather see one focused extra analysis (page 4), and you can spend up to 20 minutes of the four hours on it. More output isn't the goal.

# The files and how they join

Every CSV has a header. Read identifiers and codes as text. Blank cells are missing values, and dates are `YYYY-MM-DD`. Nobody here is a real person, so report exact counts and don't suppress small cells.

| File in `data/` | One row is / columns |
|---|---|
| `patients.csv` | One shared patient: `patient_id`, `index_date`, `age_at_index`, `death_date`. |
| `patient_linkage.csv` | One exact match: `patient_id`, `claims_id`, `ehr_id`. |
| `claims_patients.csv` | One claims patient: `claims_id`, `observation_start`, `observation_end`, `part_d`. |
| `ehr_patients.csv` | One EHR patient: `ehr_id`, `observation_start`, `observation_end`, `baseline_psa_ng_ml`, `grade_group`. |
| `claims_events.csv` | One claims event row: `claims_id`, `event_date`, `code`. Repeats are possible. |
| `ehr_events.csv` | One EHR event row: `ehr_id`, `event_date`, `code`. Repeats are possible. |
| `code_map.csv` | One code within a source: `source`, `code`, `event_type`, `modality`. |

## Shared patient information

The roster is men with an already confirmed first prostate-cancer diagnosis. `index_date` is the date of that diagnosis, so use it as given and don't rebuild it from event codes. `age_at_index` is age in completed years, already worked out. There's no separate step to confirm incident diagnosis.

`death_date` comes from a shared synthetic registry, and both sources use the same date. Assume complete death ascertainment (every death that occurred has been recorded) through 2023-12-31. A blank means no death was recorded by then. A source's recorded `observation_end` can fall after the death date (an administrative artefact, since records aren't always closed at death). Clinical observation can't continue beyond death, so truncate each span at the death date before you check eligibility.

## Linkage and observation

The crosswalk is complete and one-to-one: each roster patient has exactly one claims ID and one EHR ID. Use it. Numeric suffixes don't tell you who is who. Before you join events, check the IDs are unique and look for unmatched ones, because an event-to-event join can multiply rows.

Each source patient table has a single continuous observation span with inclusive endpoints. The claims spans already stand for qualifying fee-for-service coverage, so there are no segments to merge and no Medicare Advantage rules to apply in this version.

`part_d` is 1 when oral drug fills can be captured and 0 when they can't. It doesn't affect eligibility, but it can change what treatment gets recorded. The EHR, for its part, can miss care delivered outside its own health system.

## EHR values and event codes

The EHR PSA value (ng/mL) and grade group (1–5) are baseline values we've supplied, covering the 90 days before diagnosis through index. You don't need to pick a nearest measurement. Treat missing grade as its own category, and don't impute PSA or grade.

Events map on source plus code, where source is `claims` or `ehr`. All the codes are invented for this exercise. Rows with `event_type = treatment` are prostatectomy, radiation or systemic therapy, and `modality` gives the category. Rows with `event_type = other` include PSA tests and must never count as treatment. Every event code has a mapping.

The linked view is simply the union of what was recorded. It isn't a third data source and it isn't a clinical gold standard.

## A question of your own

The required outputs are a starting point. We'd strongly like to see one more analysis or visual of your own, because it shows us how you reason from data. Pick a question the two required plots don't already answer. The comparison, method and display are up to you. There's no prescribed extension and no expected finding.

Say why the question matters and use a clear denominator. Keep what you observed apart from what might explain it, and think about uncertainty and at least one alternative reading. A negative or inconclusive result is fine. One thoughtful comparison is plenty. Don't go past four hours, and don't trade away a correct core analysis for a complex model.

# Analysis rules and required results

This section says who goes into each group, what counts as treatment, and what to report. Everything is measured from the **index date**, the day of first confirmed diagnosis, which is day 0. The time before it is the **look-back** (pre-index observation used to confirm that each source was capturing care before diagnosis) and the 365 days after it are the **follow-up** (the post-index period in which treatment and death are ascertained).

## 1. Build two nested groups

The two questions need two cohorts because they rely on different data. Death is ascertained from the shared registry independently of claims or EHR observation, so the mortality cohort only requires pre-index observation. Treatment capture is stricter. Both sources must observe the patient throughout days 0–365, because a source can only record treatment delivered during its observation period. Without this rule, missing data would be misclassified as no treatment (a patient lost to observation would look untreated).

Start from the full roster and report how many patients remain and are excluded after each step.

| Step | Rule | Result |
|---|---|---|
| 1. Age | Age at index is 66 or older. | |
| 2. Look-back | In **both** sources, observation starts on or before `index_date` − 365 days, and the observation span includes the index date. | **Baseline cohort**, used for the death endpoint (page 7). |
| 3. Follow-up | In **both** sources, observation continues to at least `index_date` + 365 days (see below). | **Capture cohort**, used for all treatment results and the pattern plot. |

**Effective end of observation.** For each source, take the earliest of three dates: the recorded end of its observation span, the patient's registry death date, and the data cutoff of 2023-12-31. That is the source's *effective end*. A recorded `observation_end` can fall after the death date (an administrative artefact: enrolment and record systems aren't always closed at death), but clinical observation can't continue after death, so the death date truncates the span. Step 3 requires both effective ends to fall on or after day 365.

Two consequences follow. A patient with a death date before day 365, or whose observation ends before day 365, is excluded from the capture cohort at step 3 but retained in the baseline cohort. A patient whose death date is exactly day 365 is retained in both, because the effective end then equals day 365 and meets the on-or-after condition.

**Worked example.** A patient is diagnosed on 2020-02-04. For step 2, each source's observation must start on or before 2019-02-04 and the span must include the index date. For step 3, day 365 is 2021-02-03, and each source's effective end must be on or after that date. The window is defined in elapsed days, not calendar years, so 29 February 2020 counts as one of the 365 days and day 365 falls one day before the anniversary.

## 2. Decide whether each patient was treated

**The window.** Count events from the index date (day 0) through day 365, both ends included. That is 366 calendar dates. A treatment on day 365 counts and one on day 366 doesn't. In the example above, 2021-02-03 counts and 2021-02-04 doesn't.

**What counts as treatment.** An event counts when `code_map.csv` labels its code as `treatment`: prostatectomy, radiation or systemic therapy. All three count equally. PSA tests are labelled `other` and never count.

**One answer per patient.** For each capture-cohort patient, create two yes/no flags: any treatment in claims, and any treatment in the EHR. The *linked* flag is yes when either of those is yes. A patient is counted once, whatever the number of duplicate rows, repeat administrations (e.g. serial ADT injections) or records in both sources.

**Results to report.** Make a table with one row for each of the three views (claims, EHR, linked). Give the number treated, the number of patients (the same for all three rows: the capture cohort), the percentage treated and a 95% Wilson interval. A library function for the interval is fine. Then report the absolute difference between the linked and claims figures in percentage points. That difference equals the proportion of patients whose treatment was recorded only in the EHR.

The three views describe the *same* patients, so the measurements are paired (each patient contributes to all three). Don't treat them as independent groups or apply a test that assumes independence.

**Agreement between sources.** In the text, place every capture-cohort patient in exactly one of four groups. Give the counts in the text (no table or test needed); they must add up to the capture-cohort size:

- treatment recorded in both sources,
- in claims only,
- in the EHR only,
- in neither.

## 3. Make two plots

**Plot A: treatment capture.** Show the claims, EHR and linked percentages side by side with their Wilson 95% intervals. Say how many patients are in the cohort, label the percentage axis and name the interval method. A bar or a point plot is fine, whichever reads better.

**Plot B: initial treatment patterns.** Using the linked records for days 0–365, assign each capture-cohort patient to one of five mutually exclusive groups:

- No recorded treatment
- Prostatectomy only
- Radiation only
- Systemic therapy only
- Multiple modalities (two or more different treatment types)

Repeated records of the same modality don't qualify a patient for Multiple modalities. Show counts or percentages with a clear denominator, and make sure the five groups add up to the cohort size.

Explain each plot in a sentence or two. Two treatments can be recorded on the same day, so these groups describe observed combinations of modalities. They don't show treatment sequence, switching, lines of therapy or treatment intent (curative or palliative). A plot of grade is no longer required, but grade is still open as an independent question.

## 4. Assess data fitness for purpose

In the capture cohort, report the median baseline PSA among valid, non-missing values, and the number and percentage of patients with PSA missing and with grade group missing. Then discuss three ways treatment can be missing:

- **Care outside the EHR's health system.** The EHR records only care delivered within its own institution.
- **Part D.** Where `part_d` is 0, pharmacy-dispensed oral agents (such as oral ADT) don't appear in claims. Injectable therapy is still captured.
- **Care in neither extract.** Some treatment is recorded in neither source, so the linked figure likely underestimates the true proportion treated.

The intervals describe sampling variation only. They don't correct for capture bias (systematically missing records), and they don't establish external validity (whether the results would hold in other populations).

# A clinical endpoint: death within 365 days

Estimate the proportion of the baseline cohort with a recorded death from any cause on days 0–365 after diagnosis, inclusive. This is a fixed-horizon, all-cause mortality endpoint. It isn't prostate-cancer-specific mortality, and it isn't a comparison of how well treatments work.

## Define it before you count

- **Population.** Age-eligible, linked patients with 365 days of prior observation in both sources. That's the cohort after the look-back step and before you ask for post-index observation through day 365.
- **Time origin.** `index_date`, not the treatment date. The horizon is `index_date` + 365 days.
- **Event.** A shared-registry `death_date` from index through the horizon, inclusive. A death on day 365 counts and one on day 366 doesn't.
- **Ascertainment.** Assume the registry records every death through 2023-12-31, and check that every baseline patient's horizon falls on or before that date. It does for all the supplied index dates. Under this assumption a blank death date means no death by the horizon, and a death after the horizon is not an event within 365 days.
- **Source observation.** If claims or EHR observation ends before the horizon, the patient still stays in this registry-based endpoint. Don't require survival to day 365 or continued source observation for inclusion (both would condition on the outcome or on follow-up).

Everyone has registry ascertainment to the same fixed horizon, so a simple proportion with a Wilson interval will do. You don't need a Kaplan–Meier curve or a censoring model. With real data you'd have to verify registry completeness rather than infer from a blank death date that the patient was alive.

## One small endpoint table

Show the endpoint name, baseline denominator, deaths within 365 days, mortality percentage and Wilson 95% interval. Keep the denominator visible, since it won't match the treatment-capture denominator. Add a brief explanation of why using the capture cohort would underestimate early mortality (it excludes, by design, patients with a death date before day 365).

Don't compare mortality across the day-0–365 treatment-pattern groups. Those groups are defined using information from later in the same window, so a comparison from diagnosis can introduce immortal-time bias (a patient must survive long enough to receive a treatment, so treated groups are guaranteed survival time) and selection bias. Adjusted survival modelling is outside this task.

## What we're looking for

Mainly, whether you can tell a clinical endpoint from a treatment-capture measure, pick the right population and time window, and recognise when a convenient denominator changes the quantity being estimated. The dates are all there. The interpretation is where the work is.

# Submission, checks and interpretation

Before you calculate anything, write a dated plan of up to 80 words, either in the report or in a small text file. Note any later changes. You don't need a Git tag.

## Check the data and keep an issue log

Check keys, joins, dates, allowed values and missingness. Save `data_issues.csv` or a text log with these fields: file, record ID, column, original value, issue, action, reason and likely impact. For expected blanks and repeated patterns, summarise them instead of logging every row.

Leave the raw files untouched. Make your cleaning decisions in code and explain them, and don't guess at replacements the data doesn't support. We're assessing your checks and your reasoning, not whether you recovered some intended correction. The log doesn't count toward the report's table or word limits.

## What to submit

1. **Code, README and data-issue log.** One command should reproduce every result from `data/`. Use relative paths and a separate output folder. List package versions and random seeds. Your usual R or Python setup is fine.
2. **At least three automated checks**, run by that same command. Cover one-to-one linkage and unmatched IDs, treatment on day 365 versus day 366, and counting each patient once when treatment is repeated or appears in both sources. Hand-built examples with stated expected results are enough.
3. **One HTML or PDF report of at most 800 narrative words.** The core outputs are three tables (cohort flow, treatment capture, mortality endpoint) and two plots (capture and treatment patterns). Give source agreement in the text. For your extension you can add one table and/or one plot, within the same word limit. Tables, captions and references don't count toward the words, but the plan and assumptions do. Export all plot files.

Stay focused. Answer the main question, interpret the plots, mention PSA/grade missingness and spell out the specific limitations. If you did an extension, state the question and what your result can and can't tell us. Treatment can still be missing from both extracts.

The Wilson intervals express sampling uncertainty (how much a proportion might vary if these patients were a random sample from a larger population). Within this finite synthetic extract the observed proportions are exact. Explain that difference in plain language, and don't present the interval as the whole of the uncertainty.

## Ground rules

- Leave the supplied CSVs as they are. Don't download other patient data, look up more codes or try probabilistic matching. The supplied crosswalk is the authority.
- You don't need survival analysis, a causal model, a dashboard or a reconstructed diagnosis. Keep the extension modest, because a sound comparison shows more judgement than a complicated model.
- You may use AI tools. Tell us which ones and what they helped with. Be ready to explain your code and make a small change to it in a debrief.
- Put the time you spent and anything unfinished in the README. We won't guess at effort from how much output there is or how fast it runs.
- `data/CHECKSUMS.sha256` holds the input file hashes if you want to check integrity. That's optional. No data-generation code or solution is in the assignment folder.

## How we'll assess it

| Area | Points |
|---|---|
| Correct joins, cohort and date rules | 25 |
| Endpoint definition, denominators and statistics | 20 |
| Plots that make the findings clear | 20 |
| Interpretation and limitations | 15 |
| Independent question and analytical inference | 10 |
| Reproducible code, checks and issue log | 10 |

The core is worth 90 points and your extension can earn 10 more. Those go to the question, the comparison and the reasoning, not to chart count or a significant result. Stop at four hours, extension included.

# Appendix: input columns

Use this as a reference while you check and join the files. The formats and ranges describe what valid values should look like, not a promise that every supplied cell is right. Blank cells are missing, never zero. Flag anything unexpected and record how you handled it in the issue log.

## patients.csv

One row per shared patient. Everyone in the roster is a man with a confirmed first prostate-cancer diagnosis, so no sex or diagnosis-confirmation field is needed.

| Column | Expected format | Meaning and use |
|---|---|---|
| `patient_id` | Text; required; unique | Shared study identifier, like P00001. Joins to `patient_linkage.patient_id`. It isn't a source event ID. |
| `index_date` | ISO date; required | Date of the first confirmed diagnosis. This is day 0 and anchors the look-back and treatment windows. |
| `age_at_index` | Nonnegative integer; years | Completed age at index, already calculated. Keep age 66 or older. Don't derive a birth date. |
| `death_date` | ISO date or blank | Shared-registry all-cause death date. It should fall on or after index and no later than the cutoff. Assume complete ascertainment through 2023-12-31 for the endpoint. Blank means no recorded death by then. |

## patient_linkage.csv

One row per exact patient match. All three columns should be unique and non-missing. The file describes a complete one-to-one relationship, so check that your actual joins keep it that way.

| Column | Expected format | Meaning and use |
|---|---|---|
| `patient_id` | Text; required; unique | References `patients.patient_id`. Attach this shared ID to source records through the crosswalk. |
| `claims_id` | Text; required; unique | References `claims_patients.claims_id` and the ID on claims events. Numeric suffixes mean nothing for linkage. |
| `ehr_id` | Text; required; unique | References `ehr_patients.ehr_id` and the ID on EHR events. Never match it to claims by position or suffix. |

The join pattern is roster → crosswalk → each source patient table. Use the matching source ID to attach the shared ID to events. Roll events up to patient-level flags before combining the source views, since joining the event tables directly can multiply rows.

The folder also has `CHECKSUMS.sha256`, a text manifest of SHA-256 hashes for the seven CSV files. It holds no patient data. A matching hash confirms the supplied bytes are intact. It says nothing about whether they're clinically or logically correct.

# Appendix: source patient columns

## claims_patients.csv

One row per claims patient, with one continuous qualifying fee-for-service observation span. There's no segment merging to do.

| Column | Expected format | Meaning and use |
|---|---|---|
| `claims_id` | Text; required; unique | Claims-specific person key. Joins to the crosswalk and to claims events. |
| `observation_start` | ISO date; required | First observable day, inclusive. It must be on or before the observation end. Use it for the 365-day look-back check. |
| `observation_end` | ISO date; required | Last day of the supplied span, inclusive. The recorded date can fall after the death date (an administrative artefact), but observation can't continue beyond death, so truncate at the death date and at the administrative cutoff (2023-12-31) before the eligibility checks. |
| `part_d` | Integer 0 or 1; required | 1 means pharmacy-dispensed oral drugs are captured, and 0 means they aren't. It doesn't prove treatment use and doesn't change eligibility. |

## ehr_patients.csv

One row per EHR patient, with one continuous observation span and the supplied baseline clinical values. PSA and grade refer to the period from index − 90 days through index, inclusive. Individual measurement dates aren't supplied, since the selection has already been done.

| Column | Expected format | Meaning and use |
|---|---|---|
| `ehr_id` | Text; required; unique | EHR-specific person key. Joins to the crosswalk and to EHR events. |
| `observation_start` | ISO date; required | First observable EHR day, inclusive. It must not come after the observation end. Use it for the common-cohort look-back check. |
| `observation_end` | ISO date; required | Last day of the EHR span, inclusive. The effective end is the earliest of this date, recorded death and 2023-12-31. |
| `baseline_psa_ng_ml` | Positive number or blank; ng/mL | Supplied baseline PSA result. Report the median among valid non-missing values, plus the missing count and percentage in the eligible cohort. |
| `grade_group` | Integer 1–5 or blank | Supplied baseline Gleason grade group. Report missingness. It may support an independent analysis, and you can keep Missing as a category where that helps. Grade alone isn't a full clinical risk classification. |

An incomplete observation span is a cohort exclusion under the stated rules, not automatically a data error. In the same way, a recorded span end after the death date is expected in these extracts (an administrative artefact) and is handled by truncation, not automatically logged as an error. Missing clinical values should stay visible. If you find unexpected non-missing values, decide explicitly how to handle them.

# Appendix: event and mapping columns

## claims_events.csv

One row per claims event. There's no unique event key. Repeated rows and several treatments on the same date can happen.

| Column | Expected format | Meaning and use |
|---|---|---|
| `claims_id` | Text; required | References a claims patient. Attach `patient_id` through the crosswalk. One person can have many event rows. |
| `event_date` | ISO date; required | Date of the recorded service or drug fill. It should fall on or before the death date, within the source's observation span, and no later than the data cutoff. Count treatment on days 0–365 only. |
| `code` | Text; required | Invented claims code. Match it to `code_map` using source = `claims` and this code. Never infer meaning from row order. |

## ehr_events.csv

One row per EHR event. Repeats can happen, and a record in both sources doesn't mean two patients.

| Column | Expected format | Meaning and use |
|---|---|---|
| `ehr_id` | Text; required | References an EHR patient. Attach the shared ID through the EHR side of the crosswalk. |
| `event_date` | ISO date; required | Date a treatment or other event was recorded. Apply the same inclusive day-0–365 window as for claims, and check temporal plausibility before you classify (for example, events after death, outside the observation span or after the data cutoff). |
| `code` | Text; required | Invented EHR code. Match using source = `ehr` plus the code. A PSA event isn't treatment. |

## code_map.csv

One code per source. The source + code pair should be unique. This is a mapping for the exercise and not clinical coding guidance.

| Column | Expected format | Meaning and use |
|---|---|---|
| `source` | Text: `claims` or `ehr` | The event extract the mapping applies to. If needed, set it from the event filename. |
| `code` | Text; required | Exact event code, matched as text together with source. Every supplied event code should have a mapping. |
| `event_type` | Text: `treatment` or `other` | Only `treatment` feeds the treatment flags. `other` includes PSA test records and must not count as treatment. |
| `modality` | `prostatectomy`, `radiation`, `systemic`; otherwise blank | Treatment category for mapped treatment rows. Blank is expected for `other` rows. All three modalities count equally for the yes/no flag. |